

values, based on the thickness *vs.* anodization voltage relation, of 66 and 130 nm.⁸ This experiment demonstrates that with a suitable tip, SECM can be used to measure thickness changes within the range of tens of nanometers to micrometers.

Conclusions

SECM can be used to measure thickness changes of thin films, both electrically conductive and insulating, during various electrochemical processes. For measurements with a mediator, thickness changes are reflected by changes in the feedback current. With different tip sizes, the measurement sensitivity ranged from 10 to 250 nm. A mediator-free method based on the measurement of resistance between tip and counterelectrode can also be applied to thickness measurements, but with a lower measurement sensitivity.

Acknowledgments

We thank F. R.-F. Fan for help in preparing the $\text{Fe}(\text{Cp})_2\text{COONa}$ and B. R. Horrocks for helpful discussions. The support of this work by INTEVEP and the Robert A. Welch Foundation is gratefully acknowledged.

Manuscript submitted Jan. 26, 1995; revised manuscript received April 4, 1995.

The University of Texas assisted in meeting the publication costs of this article.

REFERENCES

1. A. J. Bard, F.-R. F. Fan, and M. V. Mirkin, in *Electroanalytical Chemistry*, Vol. 18, A. J. Bard, Editor, pp. 243-373, Marcel Dekker, Inc., New York (1994).
2. M. Arca, A. J. Bard, B. R. Horrocks, T. C. Richards, and D. A. Treichel, *The Analyst*, **119**, 719 (1994).
3. A. J. Bard, F.-R. F. Fan, D. T. Pierce, P. R. Unwin, D. O. Wipf, and F. Zhou, *Science*, **254**, 68 (1991).
4. A. J. Bard and J. Kwak, *Anal. Chem.*, **61**, 1221 (1989).
5. J. V. Macpherson and P. R. Unwin, *J. Phys. Chem.*, **98**, 1704 (1994).
6. J. V. Macpherson and P. R. Unwin, *ibid.*, **98**, 11764 (1994).
7. B. R. Horrocks, D. Schmidtke, A. Heller, and A. J. Bard, *Anal. Chem.*, **65**, 3605 (1993).
8. E. Gaul, *J. Chem. Educ.*, **70**, 176 (1993).
9. M. V. Mirkin, F.-R. F. Fan, and A. J. Bard, *J. Electroanal. Chem. Interfacial Electrochem.*, **328**, 47 (1992).
10. M. V. Mirkin, M. Arca, and A. J. Bard, *J. Phys. Chem.*, **97**, 10790 (1993).
11. *CRC Handbook of Chemistry and Physics*, Vol. 35, R. C. Weast, Editor, CRC Press, Boca Raton, FL (1986).

A New Class of Electrochemically and Thermally Stable Lithium Salts for Lithium Battery Electrolytes

I. Synthesis and Properties of Lithium bis[1,2-benzenediolato(2-)-O,O']borate

J. Barthel, M. Wühr, R. Buestrich, and H. J. Gores*

Institut für Theoretische und Physikalische Chemie der Universität Regensburg, D-93040 Regensburg, Germany

ABSTRACT

Disclosed is a new class of nontoxic thermally, chemically, and electrochemically stable, inexpensive lithium salts based on a chelate complex anion of boron with aromatic or aliphatic diols or carboxylic acids. The synthesis, purification, and analysis of the first member of this class, lithium bis[1,2-benzenediolato(2-)-O,O']borate ($\text{Li}[\text{B}(\text{C}_6\text{H}_4\text{O}_2)_2]$) is described and some results are given from electrochemical experiments of its solution in various aprotic solvents. The voltage window of $\text{Li}[\text{B}(\text{C}_6\text{H}_4\text{O}_2)_2]$ based solutions is limited by the oxidation of the borate at about 3.6 V *vs.* lithium. Lithium can be cycled in solutions of lithium bis[1,2-benzenediolato(2-)-O,O']borate based on different aprotic solvents. Cycling efficiencies depend strongly on the solvents used, but scarcely on contact times of the solution with lithium, or on the use of mixed electrolytes ($\text{Li}[\text{B}(\text{C}_6\text{H}_4\text{O}_2)_2]/[\text{N}(\text{CH}_3)_4][\text{B}(\text{C}_6\text{H}_4\text{O}_2)_2]$).

Introduction

The development of commercially viable secondary lithium batteries, based on a lithium metal anode and a lithium ion intercalating cathode material (*e.g.*, $\text{Li}/\text{Li}_x\text{TiS}_2$) or on two lithium ion intercalating materials (so-called rocking-chair cells¹ or lithium ion²⁻⁴ cells, *e.g.*, $\text{Li}_x\text{C}_6/\text{Li}_x\text{CoO}_2$) depends on the availability of nontoxic thermally, chemically, and electrochemically stable lithium salts.^{5,6} Further requirements are sufficiently high solubility of the lithium salt ($>1\text{M}$) in dipolar aprotic solvents (*e.g.*, PC or EC), sufficiently high conductivity of the electrolyte solution ($>5\text{ mS/cm}$), and compatibility with all cell materials.

Typical electrolytes used in early studies include coordinatively saturated large anions such as ClO_4^- , BF_4^- , PF_6^- , and AsF_6^- , which are not easily oxidized or reduced at electrodes and hence are used as the anions of the well-known lithium or tetraalkylammonium supporting electrolytes for electrochemical studies. Some problems associated with these anions have stimulated the search for substitutes.⁷ Lithium perchlorate solutions are thermally unstable and show explosion risks, especially in ethers.⁸ Lithium hexafluorophosphate itself is thermally unstable in the solid

state (decomposition temperature, 30°C) and decomposes to yield scarcely soluble LiF and the Lewis acid PF_5 .^{9,10} Lewis acids initiate the polymerization of solvents such as cyclic ethers and therefore degrade the solution, if this process is not inhibited by additives.^{11,12}

Some other Lewis-acid-based anions such as the hexafluoroarsenate and the tetrafluoroborate ion are thermally much more stable^{13,14} but show other drawbacks. The hexafluoroarsenate, which itself is of low toxicity shows environmental risks, caused by its reduction products. Lithium tetrafluoroborate often yields poorly conducting solutions, bad cycling efficiencies, and causes polymerization with cyclic ethers, just as the hexafluorophosphate.

The high lattice energy of the decomposition product LiF and its low solubility in all solvents is a common reason for stability problems with all lithium salts containing Lewis-acid-based anions. In addition, all these anions are thermodynamically unstable with lithium, *e.g.*, LiAsF_6 generates about 1600 kJ/mol heat upon reduction.¹⁵

For a couple of years the search for suitable substitutes has not been very successful. Lithium salts, based on anions like triflate CF_3SO_3^- , other perfluoroalkyl or perfluoroaryl sulfonates,⁷ closoboranes ($\text{B}_{10}\text{Cl}_{10}$, $\text{B}_{12}\text{Cl}_{12}$),¹⁶ or organoborates such as BMe_4^- ¹⁷ show no stability problems

* Electrochemical Society Active Member.

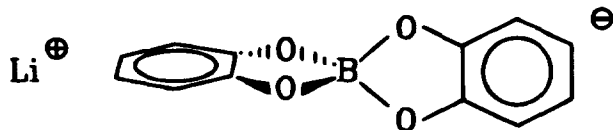


Fig. 1. Lithium bis[1,2-benzenediolato(2-)-O,O']borate.

of this type, but generally yield lower conductivities or are sensitive toward oxidation (BMe_4^-). Organoborate ions of the type BR_4^- , where R is the perfluorinated alkyl or aryl¹⁸ show increased stability *vs.* oxidation when compared with nonfluorinated organoborates,¹⁹ but cannot be easily synthesized.

Recently new lithium salts with a choice of anions based on common principles of solution chemistry have been developed, the bis(trifluoromethylsulfonyl)imide and the tris(trifluoromethylsulfonyl) methide. These molecular anions with covalent bonds, show higher chemical stability than the Lewis-acid-based anions, larger radii and hence lower lattice energies leading to good solubility, low polarizability, especially when perfluorinated, due to the low polarizability of fluorine. They show low anion solvent interaction, and generally their large charge delocalization results in an excellent electrochemical stability and a weak anion-lithium ion interaction entailing higher conductivities, despite the lower mobility of the larger anion.

These new lithium salts exhibit good safety characteristics and high thermal stability. For example, the lithium methide is stable up to 340°C²⁰ a very high temperature, when compared to lithium hexafluorophosphate (30°C).⁹

The imide is already commercially available; a pilot plant for the production of the methide is projected.⁷

Such salts are also useful electrolytes for solid polymer electrolytes.²⁰⁻²²

Chelate Complexes of the Boron, New Anions for Lithium Salts

A new class of thermally stable lithium electrolytes has recently been discovered in our laboratory,²³ with anions generally composed of a chelate complex of boron with aromatic or aliphatic diols (*e.g.*, catechol, *cf.*, Fig. 1) or carboxylic acids (*e.g.*, salicylic acid). The starting materials of the synthesis are inexpensive and of low toxicity.

Figure 1 displays several features of the new anion in common with those of the previously discussed anions: covalent bonds, large radius, and a pronounced charge delocalization due to aromatic substituents. As a consequence of the large anionic radius and charge delocalization, the solubility of the new lithium salt in various dipolar aprotic solvents and some ethers is more than 1 mol/kg, in contrast to the well-known lithium tetramethanolatoborate, which is almost insoluble in any aprotic solvent.

Despite the increased ionic radius an increase in conductivity can be expected, because ion pair formation is suppressed due to the charge delocalization of the anion. The characteristics of the charge delocalization can be seen from semiempirical quantum-mechanical calculations with the help of MOPAC²⁴ (AM1 and MNDO), showing that the mean charge at the oxygen atoms is -0.402 (MNDO) or -0.367 (AM1) for the tetramethanolatoborate anion whereas it is only -0.246 (MNDO) or -0.244 (AM1) for the bis[1,2-benzenediolato(2-)-O,O']borate; all values are fractions of the elementary charge. The charge delocalization may be increased further by fluorination of the anions. For example the bis[tetrafluor-1,2[benzenediolato(2-)-O,O'] borate shows only mean oxygen charges of -0.216 (MNDO) or -0.215 (AM1), respectively.

The effect of anionic charge delocalization on ion association may be estimated from a recent conductivity study on lithium acetates in DMSO. The association constant decreases by a factor of about 50 when the acetate ion is compared to the trifluoroacetate ion.⁶ This behavior can be understood from semiempirical quantum-mechanical calculations and electrostatic theories of the association pro-

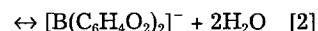
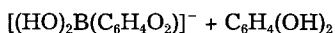
cess, because the association constant mainly depends on the charge of the oxygen atoms in the acetates. A plot of the logarithm of the association constant *vs.* the oxygen charge obtained from MNDO calculations is linear. In solvents of low dielectric permittivity this effect will cause a strong increase in conductivity of the electrolyte solution. We refer to this feature in a separate study.

The charge delocalization also affects the electrochemical stability *vs.* oxidation. This follows from semiempirical quantum-mechanical calculations showing that HOMO energies and oxidation potentials are linearly related.²⁵ The slope of this linear relation is about -1 V/eV. Recent MOPAC²⁴ (AM1 and MNDO) calculations show that the anodic stability limit can be extended by about 1.5 V by the fluorination, *i.e.*, for the bis[tetrafluor-1,2-benzenediolato(2-)-O,O']borate.

In addition to the benefits already discussed, the new class of lithium salts based on anions, which are chelates of various possible bidentate ligands shows a new feature which is essential for their thermal stability in the solid state and their stability in solutions *vs.* nucleophilic displacements of the ligand. It is known from the literature that chelates of bidentate ligands such as the benzenediol are much more stable when compared with anions based on monodentate ligands such as the methanolate. For example, the decomposition of lithium tetramethanolatoborate starts at 50°C,²⁶ whereas lithium bis[1,2-benzenediolato(2-)-O,O']borate is thermally stable up to >290°C. At 260°C the decomposition pressure of lithium tetramethanolatoborate reaches already 96.5 Torr.²⁶

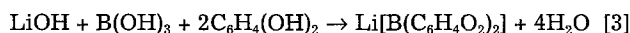
In the actual paper we report on the synthesis, analysis, and purification of the lithium bis[1,2-benzenediolato(2-)-O,O']borate, its electrochemical properties and on the synthesis, purification, and analysis of the tetramethylammonium bis[1,2-benzenediolato(2-)-O,O']borate, which was used in mixed electrolytes to increase the conductivity of the solutions. In following papers we present the synthesis and the properties of other lithium salts including those with fluorinated anions.

Synthesis and purification of the salts.—The borate anion $\text{B}(\text{OH})_4^-$ reacts with 1,2-benzenediol in a two step equilibrium reaction to yield the borate $[\text{B}(\text{C}_6\text{H}_4\text{O}_2)_2]^-$ anion in aqueous solutions



The equilibrium constant for the first step is $2.0 \times 10^4 \text{ mol}^{-1} \times \text{dm}^3$, for the second step it is very small, only $5 \text{ mol}^{-1} \times \text{dm}^3$.²⁷

Therefore it could be supposed that lithium bis[1,2-benzenediolato(2-)-O,O']borate, is obtained when LiOH, boric acid $[\text{B}(\text{OH})_3]$, and catechol are dissolved in water in stoichiometric proportions at high concentrations of the reactants in order to precipitate the less soluble lithium salt according to

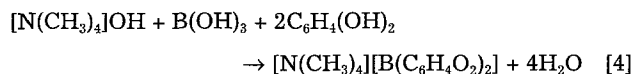


The assumption turned out to be essentially correct. After various reactions performed at different temperatures and concentrations, we performed the following synthesis.

Five hundred grams of catechol (Merck, synthesis grade), 95.27 g LiOH · H₂O, (Aldrich, purity >99.9%) and 140.31 g boric acid (Merck, p.a. grade) were transferred to a Pyrex flask fitted with a reflux condenser. The compounds were rinsed with high-purity argon (99.999%). 250 ml of purified water saturated with argon were added. The inert atmosphere is necessary because the catechol is extremely sensitive to oxygen. The flask was slowly heated to 105°C where a clear brownish solution was obtained. The solution was cooled down to 50°C where the beginning crystallization of the lithium salt was observed. Colorless plates of $\text{Li}[\text{B}(\text{C}_6\text{H}_4\text{O}_2)_2] \cdot 2\text{H}_2\text{O}$ were obtained after subsequent crystallization at room temperature (14 h) and at 3°C (48 h)

and a final filtration under inert gas. The hydrate was dried at increasing temperatures from 0°C (4 h) to 25°C (5 h) and finally to 115°C (50 h) under reduced pressure ($<10^{-2}$ Torr) yielding 323.9 g of the lithium salt (yield: 61.1%). The raw lithium salt was dissolved in 800 ml boiling acetonitrile (Baker, HPLC grade) refluxed under inert gas and then slowly cooled to -10°C, where crystals were obtained. The recrystallization steps were repeated with decreasing amounts of AN (600, 500, and 280 ml); the finally obtained AN solution is colorless. The colorless solvate crystals were dried for 6 h at 110°C and for 50 h at 160°C under reduced pressure up to weight constancy, yielding a white powder. The mean yield for the purified lithium salt is 72% per recrystallization step.

Similar to Eq. 3, an aqueous solution of tetramethylammonium hydroxide reacts with boric acid and catechol to yield the tetraalkylammonium salt



The optimized synthesis is as follows: 86.98 g catechol (Merck, synthesis grade), 24.43 g boric acid (Merck, p.a.), 360 ml 10% aqueous solution of tetramethylammonium hydroxide (Merck, polarographic grade), and 300 ml water were heated to about 80°C under purified argon in a Pyrex flask fitted with a reflux condenser. In order to initiate the crystallization of the salt, some previously prepared crystals were given to the yellowish brown hot solution. This step is essential for the success of the synthesis. Its omission results in a noncrystallizable oil. The remaining solution was separated from the crystals, which are dried under reduced pressure at 60 to 70°C with a small loss in weight (0.3%), because no hydrate is formed. The yield of the raw material is 56.6%.

The previously used crystals are obtained by heating a small fraction of the solution with the help of a heating fan, and cooling the solution to 0°C where crystallization is induced by scratching with a glass rod.

Recrystallization is performed from acetone (Merck, p.a.) (50 g of the raw material in 125 ml acetone) at 50 to 55°C under inert gas. The recrystallization is repeated four times in order to obtain a colorless solution and finally colorless crystals, which are dried at room temperature at reduced pressure and finally at 50 to 70°C, where a colorless powder is obtained. The yield per crystallization step from acetone is 87.9%.

Analysis of the salts.—Reversion of Eq. 3 and 4 corresponds to the hydrolysis of the borates, which is the basis for their potentiometric titration with aqueous HCl at elevated temperatures (50°C), where the reaction is sufficiently fast. The boron content of the nonpurified lithium salt obtained from aqueous solution after the decomposition of the hydrate is 4.611%, the purified product contains 4.622% boron in excellent agreement with the theoretical boron content of 4.621%.

The purified tetraalkylammonium salt shows a boron content of 3.580%, the theoretical value is 3.590%.

The 250 MHz ^1H -NMR spectrum (Bruker AC-250) of a 0.3M solution of the lithium salt in DMSO- D_6 shows a singlet at 6.48 ppm (TMS). This is a spectrum of the type AA'BB' with equal chemical shifts, proved by the equivalent result for the catechol dimethyl ether. The purified product shows no signals corresponding to the solvent AN in ^1H -NMR spectra, which is observable for the solvate at 2.0 ppm. A coulometric Karl-Fischer-titration (Abimed CA-02) of the solvent THF and of a THF-based solution of the lithium salt yields a water content of 56 ppm, i.e., in the usual range of purified lithium salts. No peaks corresponding to the starting material catechol are observed for the lithium salt.

The 250 MHz ^1H -NMR spectrum (Bruker AC-250) of a 0.5M solution of the tetraalkylammonium salt in DMSO- D_6 shows a singlet at 6.52 ppm (TMS) which can be attributed to the protons of the borate-anion and a singlet at 3.03 ppm due to the protons of the methyl groups, where the ratio of

the integrals corresponds to the theoretically expected value.

^{13}C -NMR-spectrum of the lithium salt exhibits three signals at 107.6, 117.3, and 151.6 ppm, the tetraalkylammonium salt approximately shows the same signals at 107.7, 117.4, and 151.5 in addition to a triplet for the methyl groups at 54.37, 54.43, and 54.49 in accordance with expectation.

The purified lithium salt decomposes at high temperatures without melting (above 290°C). When exposed to fire, it carbonizes coloring the flame green and red yielding a glass finally in contrast to the catechol which melts and burns completely.

A 0.9 molal solution of the lithium salt in the mixed purified solvent PC/EC [60 weight percent (w/o) PC] was transmitted to a flask, containing a piece of freshly cut lithium, evacuated and sealed. No visual change was observed, even after a storage of 3 months at 80°C.

The purified lithium salt shows a high solubility in THF, 2-Me-THF, DME, and PC whereas it is scarcely soluble in dioxolane, diethylether, or toluene. The tetraalkylammonium salt is soluble in PC, dioxolane, AN, but scarcely soluble in DME and THF and nearly insoluble in toluene.

In addition to these observations proving the purity and some expected properties of the salts, the analysis of the lithium salt solvates, performed gravimetrically and by NMR spectroscopy, gives an indirect hint of the existence of the salt itself. Gravimetric determinations yield 2.22 mole water per mole of the nonpurified lithium salt hydrate and 2.01 mole AN per mole of the purified lithium salt solvate. From NMR investigations 2.13 mole water or 1.78 mole AN are obtained.

Lithium salt solutions in various nonaqueous solvents show specific conductivities κ in the expected range of some mS/cm and the usual conductance increase upon addition of a cosolvent, cf., Table I, where some preliminary values are given, sufficient to exemplify the expected behavior^{5,6} and useful for planning electrochemical measurements.

Purification of solvents and preparation of solutions.—The purification of the solvents was performed according to proven literature procedures. The final distillation for every solvent was carried out at reduced pressure or under purified argon with distillation columns rated at about 50 theoretical plates. All purified solvents and the salt solutions were stored under purified argon in especially constructed glass vessels desiccated at elevated temperatures under purified argon. The preparation of solutions and electrochemical cells was performed in a glove box (MBraun MB 150 B G) with the nitrogen removing equipment (MBraun MB 200N) under purified argon at low oxygen (<1 ppm), nitrogen (<1 ppm), and water (<1 ppm) contents.

Propylene carbonate (PC), (Fluka, purum >99%) shows no organic impurities in gas chromatograms after the purification according to Fujinaga and Izutsu.²⁸ The water content was about 12 ppm.

Ethylene carbonate (EC), (Fluka, purissimum, >99%) purified by refluxation over CaH_2 and subsequent distillation at 0.5 Torr shows a purity of 99.994% (GC), and a water content of less than 20 ppm.

Tetrahydrofuran (THF), (Merck, p.a.) was dried on a column with activated alumina (Woelm, neutral) under argon, refluxed over LiAlH_4 for 24 h and finally distilled,

Table I. Conductivities of some solutions of the new electrolyte.

Solution ($X = [\text{B}(\text{C}_6\text{H}_4\text{O}_2)_2]$)	κ (mS/cm)
1.231 m LiX in PC (27°C)	0.6
0.899 m LiX in DME (26°C)	1.7
1.11 m LiX in PC/DME (45.7 w/o PC) (25°C)	3.0
0.4 m Me ₄ NX and 0.555 m LiX in PC/DME (52.8 w/o PC) (23°C)	5.5
0.7 m LiX in EC/DME (51 w/o EC) (25°C)	5.6

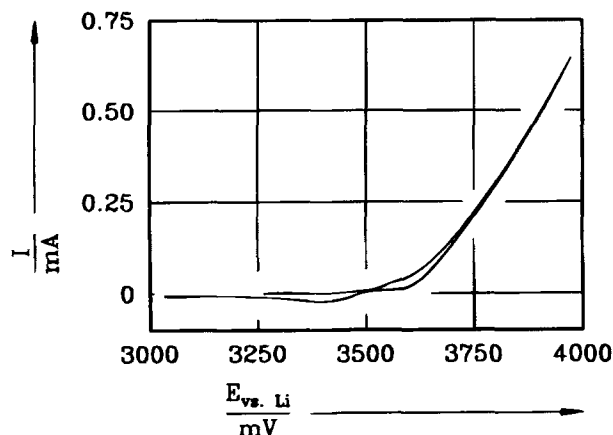


Fig. 2. CV of 1.2 M lithium bis[1,2-benzenediolato(2-)-O,O']borate/PC solution at a stainless steel electrode showing the anodic stability of the anion.

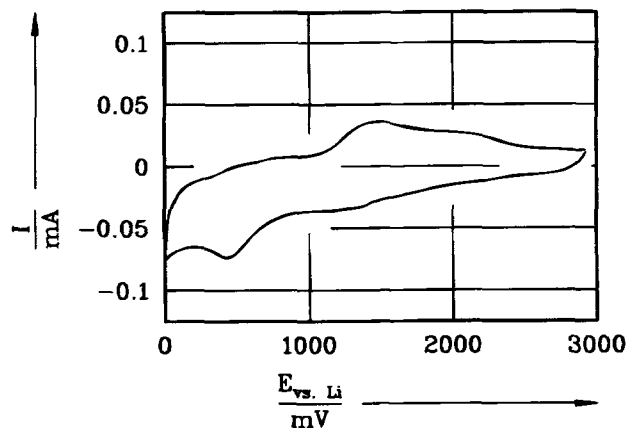


Fig. 3. Residual current of a 1.2 M $\text{Li}[\text{B}(\text{C}_6\text{H}_4\text{O}_2)_2]/\text{PC}$ solution, as obtained from the second sweep of a CV.

yielding about 20 ppm organic impurities (GC) and 11 ppm water.

The purification of 2-Me-THF has already been described in this journal.²⁹ The solvent shows <20 ppm organic impurities (GC) and its water content is smaller than 10 ppm.

All chromatographic investigations were performed with a Perkin-Elmer gas chromatograph PE 8302B with flame ionization detector (FID) equipped with a (OV1), (CPMS), or (PEG) column, dimethylpolysiloxane (Ohio Valley) (OV-1), 7% cyanopropyl-, 7% phenyl-, 86% methylsilicone (CPMS), or poly(ethyleneglycol), carbowax (PEG) column.

The electrochemical equipment.—The electrochemical experiments were carried out at 25°C in thermostated glass cells with a volume of 10 to 20 ml, equipped with three electrodes; for a description see Ref. 29. The working electrode was a stainless steel (SS 304) rod, area of surface 0.503 cm² or a gold rod, area 0.0707 cm². The working electrodes were polished with alumina, rinsed with water and acetone, and dried before use. Counter- and reference electrodes were made of lithium foils (thickness 1.5 mm Chemetall, >99.95%).

The cyclic voltammetry (CV) experiments were performed at a sweep rate of 20 mV/s in the potentiostatic mode using an IM-5d electrochemical measurement system. (Zahner-Elektrok, Kronach, Germany).

Galvanostatic cycling experiments were controlled with a Bank Wenking POS 73 potentiostat in the potential following mode. Two types of galvanostatic lithium cycling experiments, GC I and GC II, were performed as described in a preceding publication.²⁹

Results and Discussion

Figure 2 exhibits the anodic stability limit of the electrolyte as obtained from a CV of a 1.2 M PC solution of lithium bis[1,2-benzenediolato(2-)-O,O']borate using a stainless steel working electrode. The sweep range covers the potential from the open-circuit potential E_R vs. lithium up to 4000 mV vs. lithium, and back to E_R .

The shape of the approximately linear part of the CV at potentials above 3600 mV is not affected by using a gold electrode instead of a stainless steel electrode. The change of solvents (PC, EC, THF, and Me-THF) changes only the slope of the lines, which may be attributed to the change of solution conductivity. As the stability limit of the used solvents is much higher than 3.6 V,³⁰ it is reasonable to assume that the borate is oxidized. Repeated (up to five times) cycling in this potential range does not change the shape of the CV, showing that no electrode passivation takes place. The anodic limit does not change upon replacement of the lithium salt by a mixture of the tetraalkylammonium salt and the lithium salt.

The residual current in the sweep range E_R to 0 V and back to E_R (all vs. Li) shows a small cathodic peak at 350 to 450 mV and a small anodic peak at about 1200 mV, see Fig. 3, which also is observed for $\text{LiClO}_4/2\text{-Me-THF}$ solutions.²⁹ The peaks cannot be attributed to electrochemical reactions of the borate anion.

According to Malik *et al.*³¹ they are presumably due to the underpotential deposition and dissolution of lithium. The shape of the curve only scarcely changes during the first sweeps, and generally reaches a steady state after two cycles. The residual current also is scarcely dependent on the contact time of the solution with lithium. Addition of the tetramethylammonium salt to the lithium salt solutions does also not alter the residual current.

All CV scans, starting point at E_R , down to -500 mV vs. lithium for lithium deposition, up to 3000 mV vs. lithium for its dissolution and back to E_R , show a similar shape for every lithium salt solution; for PC as the solvent, cf. Fig. 4, they coincide during the first observed 5 scans, for other salt-solvent mixtures they change only scarcely during cycling.

A shallow low capacity cycling efficiency can be calculated from the CVs. It is strongly dependent on the solvent and ranges from about 90% for PC-based solutions to only 5% for THF-based solutions of $\text{Li}[\text{B}(\text{C}_6\text{H}_4\text{O}_2)_2]$, see Table II. Mixed electrolytes $\text{Me}_4\text{N}[\text{B}(\text{C}_6\text{H}_4\text{O}_2)_2]/\text{Li}[\text{B}(\text{C}_6\text{H}_4\text{O}_2)_2]$ do not change the cycling behavior of electrolyte solutions, if based on the same solvent, cf., Table II.

Table III shows that the cycling efficiency is scarcely dependent on contact time of the solution with lithium before deposition.

In GCI experiments 100 mC/cm² of lithium was plated (1 mA/cm²) onto a stainless steel electrode and then cycled at strip and plate currents of 1 mA/cm² (anodic limit

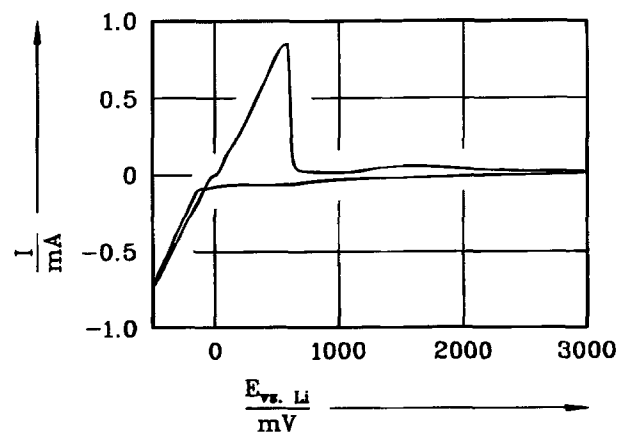


Fig. 4. CV of a 1.1 M $\text{Li}[\text{B}(\text{C}_6\text{H}_4\text{O}_2)_2]/\text{PC}$ solution, showing electrodeposition and electrodisolution of lithium.

Table II. Influence of electrolyte solutions on cycling efficiency.

Electrolyte solution (X = [B(C ₆ H ₄ O ₂) ₂])	Cycling efficiency (%)
1.231 m LiX in PC	94.7
0.899 m LiX in DME	15.3
1.184 m LiX in THF	5.2
0.595 m LiX in 2-Me-THF	12.3
1.110 m LiX in PC/DME, 54.3 w/o DME	82.2
0.519 m Me ₄ NX, 0.293 m LiX, PC	95.7

2000 mV vs. lithium) up to 100 times. The mean coulombic efficiency was about 80% for PC solutions during the first 20 cycles, decreasing thereafter to very low values (mean coulombic efficiency about 31% for 100 cycles). In ether-based solutions the efficiency is generally low (DME, 26%; THF, 22%; 2-Me-THF, 20%) and is approximately constant with the exception of the first few cycles. Addition of the tetraalkylammonium salt to Li[B(C₆H₄O₂)₂] solutions did not change the cycling behavior.

At glassy carbon electrodes (Metrohm, 0.0707 cm²) efficiencies increase continuously during the first 1100 cycles reaching finally an efficiency of about 90%. It is interesting to note that the electrolyte solution (0.473 m Li[B(C₆H₄O₂)₂] and 0.896 m [N(CH₃)₄][B(C₆H₄O₂)₂] in a mixture of PC and DME, 56.9 w/o PC) did not degrade. Neither a coloration nor development of gas bubbles were observed. We therefore suppose that the low efficiencies at the beginning of the cycling were caused by the development of a protecting layer.

In GCII experiments starting with a preplated amount of lithium (4 C/cm², 1 mA/cm²), at plate and strip currents of 1 mA/cm² and cycles of 25% DOD we observed generally low efficiencies (e.g., PC, 18%; EC/DME 51 w/o EC, 12%). The only exception was an optimized solution (EC/PC 64 w/o EC, 0.9 m), where an efficiency of 80% was observed, i.e., in the range of solutions, e.g., of EC/PC 50 w/o EC, 0.5 m lithium bis(trifluoromethylsulfonyl) imide (64%).

Summary and Conclusion

A new class of nontoxic thermally, chemically, and electrochemically stable inexpensive lithium salts based on a chelate complex of the boron with aromatic or aliphatic diols or carboxylic acids has been disclosed.

The synthesis, purification, and analysis of a member of this class, lithium bis[1,2-benzenediolato(2-)-O,O']borate is described, and some results are given from electrochemical experiments of its solutions in aprotic solvents. The voltage window of borate-based solutions is limited by the oxidation of the borate at about 3.6 V vs. lithium, independent of the solvent. Cycling efficiencies depend strongly on the solvents used, but scarcely on the contact times of the solution with lithium.

The voltage window is too small for its application in lithium ion rechargeable batteries with strongly oxidizing cathode materials such as manganese oxide spinels, requiring an electrochemical stability range of about 4 to 4.5 V. In order to increase the voltage window and the conductivity of the solutions, other members of this new class of electrolytes, including fluorinated analogues of the described borate are investigated.

Table III. Influence of contact time of 1.231 molal lithium bis[1,2-benzenediolato(2-)-O,O']borate solution in PC as the solvent with lithium on cycling efficiency.

Contact time (h)	Cycling efficiency (%)
0	94.7
1	92.6
8	91.8
13	91.0

Manuscript received Feb. 13, 1995.

University of Regensburg assisted in meeting the publication costs of this article.

REFERENCES

1. B. Scrosati, *This Journal*, **139**, 2776 (1992).
2. T. Nagaura and K. Tozawa, *Prog. Batt. Solar Cells*, **9**, 209 (1990).
3. T. Nagaura, *ibid.*, **10**, 218 (1991).
4. K. Sekai, H. Azuma, A. Omaru, S. Fujita, H. Imoto, T. Enolo, K. Yamaura, M. Yokogawa, and Y. Nishi, Extended Abstract, p. 108, 6th International Meeting on Lithium Batteries, Münster, Germany (1992).
5. J. Barthel and H. J. Gores, *Solution Chemistry: A Cutting Edge in Modern Electrochemical Technology in Chemistry of Nonaqueous Solutions*, G. Mamantov and A. I. Popov, Editors, Chap. 1, p. 1, VCH, New York (1994).
6. H. J. Gores and J. Barthel, *Pure Appl. Chem.*, In press.
7. L. A. Dominey, in *Lithium Batteries*, G. Pistoia, Editor, Elsevier, Amsterdam (1994).
8. G. H. Newman, R. W. Francis, L. H. Gaines, and B. M. L. Rao, *This Journal*, **127**, 2025 (1980).
9. T. C. Ehlert and M. M. Hsia, *J. Chem. Eng. Data*, **17**, 18 (1972).
10. J. R. Waason and D. K. Hoffmann, in *Proceedings of Symposium on Power Sources for Biomedical Implantable Applications and Ambient Temperature Lithium Batteries*, B. G. Owens and N. Margalit, Editors, PV 80-4, p. 309, The Electrochemical Society Proceedings Series, Princeton, NJ (1980).
11. S. Gorin, L. Monniere, and J. Laforest, *Bull. Soc. Chim. France*, No. 4, 1265 (1967).
12. P. Dreyfuss, *Poly(tetrahydrofuran)*, Gordon and Breach, New York (1982).
13. E. W. Lawless, C. J. W. Wiegand, Y. Mizumoto, and C. Weis, *Inorg. Chem.*, **10**, 1084 (1971).
14. V. N. Plakhotnik, V. B. Tul'chinskii, and V. K. Steba, *Russ. J. Inorg. Chem.*, **21**, 451 (1976).
15. K. M. Abraham, *Electrochim. Acta*, **38**, 1233 (1993).
16. J. W. Johnson and J. F. Brody, *This Journal*, **129**, 2213 (1982).
17. L. P. Klemann and G. H. Newman, *ibid.*, **128**, 13 (1981).
18. F. Kita, A. Kawakami, T. Sonoda, and H. Kobayashi, in *New Sealed Rechargeable Batteries and Supercapacitors*, B. M. Barnett, E. Dowgiallo, G. Halpert, Y. Matsuda, and Z.-I. Takehara, Editors, PV 93-24, p. 321, The Electrochemical Society Proceedings Series, Pennington, NJ (1993).
19. H. R. Horowitz, J. I. Haberman, L. P. Klemann, G. H. J. Newman, E. L. Stogryn, and T. A. Whitney, in *Proceedings of the Symposium on Lithium Batteries*, H.-V. Venkatesetty, Editor, PV 81-4, p. 131, The Electrochemical Society Proceedings Series, Pennington, NJ (1981).
20. L. A. Dominey, V. R. Koch, and T. J. Blakley, *Electrochim. Acta*, **37**, 1551 (1992).
21. M. Armand, W. Gorecki, and R. Andréani, *Second International Symposium on Polymer Electrolytes*, B. Scrosati, Editor, p. 91, Elsevier Applied Science, London (1990).
22. L. A. Dominey, T. J. Blakley, and V. R. Koch, in *Proceedings of 25th Intersociety Energy Conversion Engineering Conference*, Vol. 3, p. 382 (1990).
23. M. Wühr, German Pat. Appl. DE 43 16 104 A 1 (1994).
24. J. J. P. Stewart, *Mopac: A General Molecular Orbital Package*, Frank J. Seiler Research Laboratory, U.S. Air Force Academy, Colorado Springs, CO (1987).
25. H. Yilmaz, E. Yurtsever, and L. Toppare, *J. Electroanal. Chem.*, **261**, 105 (1989).
26. H. Steinberg, *Organoboron Chemistry*, Chap. 14, p. 618, Wiley, New York (1964).
27. K. Yoshino, M. Kotaka, M. Okamoto, and H. Kakihana, *Bull. Chem. Soc. Jpn.*, **52**, 3005 (1979).
28. T. Fujinaga and K. Izutsu, *Pure Appl. Chem.*, **1**, 97 (1974).
29. J. Barthel, M. Wühr, W. Sauerer, and B. Hefer, *This Journal*, **140**, 6 (1993).
30. S. A. Campbell, C. Bowes, and R. S. McMillan, *J. Electroanal. Chem.*, **284**, 195 (1990).
31. Y. Malik, D. Aurbach, P. Dan, and A. Meitav, *ibid.*, **282**, 73 (1990).